Auteur: Luca Letroye
Studierichting: Informatica
Oefeningenreeks 6D nr. 31.26

$$\sum_{n=1}^{\infty} \frac{n+1}{n^3+3n^2}$$

$\Rightarrow$ verhoudingscriterium

$$\frac{\frac{n+1}{(n^3+3n^2}}{\frac{1}{n}} = \frac{n+1}{n^3+3n^2} \cdot n^2$$

$$= \frac{n^3+n^2}{n^3+3n^2} = \frac{1+\frac{1}{n}}{1+\frac{3}{n}}$$

$$\Rightarrow \lim_{n\to+\infty} \frac{1+\frac{1}{n}}{1+\frac{3}{n}} = 1 \Rightarrow \text{zelfde convergentie gedrag}$$

$$\Rightarrow \sum_{n=2}^{\infty} \frac{1}{n^2} \text{ convergent} \Rightarrow \sum_{n=1}^{\infty} \frac{n+1}{n^3+3n^2} \text{ convergent}$$

References